Md Ahsanul Haque, Ph.D.
MAT-120

Final Exam

Date: December 23, 2025

Name

DISTRIBUTION OF MARKS		
Question No.	**Full Marks**	**Obtained Marks**
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
Total	50	

Instructions:

- Write your full name (first **and** last name) **inside** the provided box.
- This assignment contains 6 problems. It is **YOUR responsibility** to check that your copy has all 6 problems.
- Make sure to read the instructions for each problem carefully.
- Answer **5** questions. If you answer 6, the first 5 will be graded.
- Use correct mathematical notation.
- Write cohesively and follow a logical sequence of steps.
- Check that your answer can be read and understood by the grader.

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1. (10 points) Evaluate the following indefinite integrals:
   (a) $\int \left(e^{2x}+1\right)^2 e^{2x}\,dx$
   (b) $\int \dfrac{1}{x(\ln x)^2}\,dx$
   (c) $\int (2x-1)e^{x^2-x}\,dx$
   (d) $\int \frac{2x}{\sqrt{x^2+9}}\,dx$
   (e) $\int \sec^2(3x)\,dx$

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2. (10 points) Use L'Hôpital's Rule to evaluate the following limits:

(a) $\lim\limits_{x\to 1} \dfrac{\ln x}{x-1}$

(b) $\lim\limits_{x\to 0} \dfrac{e^x - 1 - x}{x^2}$

(c) $\lim_{x\to 0} \frac{e^x-1}{x}$

(d) $\lim_{x\to 0} \frac{1-\cos x}{x^2}$

(e) $\lim_{x\to \infty} \frac{x^2}{e^x}$

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3. (10 points) Consider the function

$$f(x) = x^3 - 6x^2 + 9x + 1$$

(a) Find the intervals where $f(x)$ is concave up and concave down.

$$g'(x) = (x-1)(x+2)^2.$$

(b) Determine where g is increasing and decreasing.

$$h(x) = x^3 - 3x^2 + 4$$

(c) Find the absolute maximum and minimum values of h on the interval $[-1, 3]$.

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4. (10 points)

(a) A particle moves along a line so that its position is given by $s(t) = t^3 - 6t^2 + 9t$.

i. Find the velocity and acceleration.

ii. Determine the times when the particle is at rest.

(b) The side length of a cube is increasing at a constant rate of 3 cm/s.

i. Write the rate of change of the volume V with respect to time t in terms of the side length s.

ii. Determine the rate at which the volume is increasing when the side length is 10 cm.

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5. (10 points) Evaluate the definite integrals:

(a) $\displaystyle\int_0^1 \ln(1+x)\,dx$

(b) $\displaystyle\int_{-2}^2 \sqrt{4-x^2}\,dx$ ***Hint:*** **Think about a function this looks like!**

(c) $\displaystyle\int_0^{\pi/2} \cos^2 x\,dx$ ***Hint:*** **Use the double angle formula**

(d) $\int_0^1 \frac{x}{x^2+1}\,dx$

(e) $\int_0^{\pi/4} \sec^2 x \tan x\,dx$

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6. (10 points)

(a) Use the Fundamental Theorem of Calculus

i. Find $\frac{d}{dx}\int_0^x \cos(t^2)\,dt$

ii. If $F(x) = \int_1^x \frac{1}{t}\,dt$, find $F'(x)$ and $F''(x)$.

(b) Use the definition of area as a limit (Riemann sums) to express the area under $f(x) = 3x$ from $x = 1$ to $x = 3$ as a limit.
(**You do not need to evaluate the limit, just set it up correctly.**)

(c) Use sigma notation to express the following sums:

i. $1^2 + 2^2 + 3^2 + \cdots + 20^2$

ii. $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{64}$

Extra Page